

Worst-Case Distance-Aware Error Bounds for Neural Networks (Supplementary Material)

TABLE 9. Expansions of important acronyms.

Acronym	Expansion
NN	Neural network
SNN	Spline neural network
KAT	Kolmogorov-Arnold theorem
KAN	Kolmogorov-Arnold network
KKAN	Kürková Kolmogorov-Arnold network
ReLU	Rectified linear unit
MLP	Multi-layer perceptron
MLP-Spline	A network that uses MLPs and Splines in its architecture
SNR-MLP	Spectrally normalized ReLU-activated MLP
RBF	Radial basis function
GP	Gaussian process
DKL	Deep kernel learning
DUE	Deterministic uncertainty estimation
SNGP	Spectral-normalized neural Gaussian process
DAREK	Distance-aware error for Kolmogorov networks
K-DAREK	distance-aware error for Kürková Kolmogorov networks
CBF	Control barrier function
MPC	Model predictive control
MSE	Mean squared error

A. Notation and Acronyms

Table 9 lists the expansions of the acronyms used throughout this paper, and Table 10 summarizes the notation used.

B. Real-Data Regression with Reduced Feature Dimensionality

We repeated the real-data regression experiment of Sec. VI using only the first two continuous input features of each dataset, rather than the full feature set. Table 11 reports the results, which are consistent with the full-feature results in Table 5: K-DAREK (KDK2, KDK3) achieves the fewest violations across datasets, while RMSE remains competitive with the strongest baseline.

C. Additional High-Dimensional Experiments:

We evaluate Table 7 with a complete dimensional sweep using 2, 5, 10, 20, 50, 100, and 200 features and two feature-extraction strategies. The first applies PCA directly to flattened $224 \times 224 \times 3$ images. The second passes each image through a pretrained ResNet-18 feature extractor, producing a $7 \times 7 \times 512$ representation, which is flattened and reduced using PCA. Table 12 reports the complete results for both representations.

DAREK and K-DAREK reach SDA values of approximately 95% with raw-image PCA and 90% with PCA-reduced ResNet-18 features. These results indicate strong but imperfect sampled distance-aware behavior and show that SDA depends on both feature dimension and representation. ResNet-18 features generally yield lower RMSE and higher IoU, despite producing slightly lower SDA than raw-image PCA. Thus, higher predictive performance does not necessarily imply higher sampled distance-awareness.

Fig. 7 shows representative KDK2 predictions using 100-dimensional PCA-reduced ResNet-18 features, including

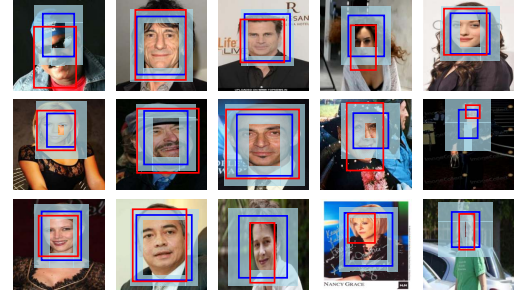


FIGURE 7. Face bounding-box prediction on sample test images from the CelebA dataset [74] using K-DAREK (KDK2). Red boxes indicate ground-truth bounding boxes and blue boxes indicate K-DAREK predictions. The shaded blue region indicates the error estimates in the x- and y-coordinates. Note that it uses only 100 projected dimensions from the 25k-dimensional feature space of ResNet-18.

both bounded and violated predictions. The observed violations may reflect underestimation of the effective Lipschitz constant in the reduced feature space, prediction error associated with dimensionality reduction, and noise in the CelebA bounding-box annotations.

TABLE 10. Notation and definitions of important quantities.

Notation	Definition	Notation	Definition
$f, \hat{f}$	Target function and trained approximator	$\hat{h}_{\text{MLP}}, \hat{h}_{\text{sp}}$	MLP block and spline block of K-DAREK
$d, d_u(\cdot)$	Input dimension $\mathbf{x} \in \mathbb{R}^d$, distance function candidate for uncertainty function u	q	Intermediate feature dimension
$\mathcal{D}$	Training dataset, $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$	L	Number of layers in each SNR-MLP
$\mathcal{X}, \mathcal{X}_{\mathcal{D}}$	Input space ($\mathcal{X} \subset \mathbb{R}^d$) and input portion of $\mathcal{D}$	τ, k	Set of knots and order of spline
$\mathbf{x}, \mathbf{x}^*$	Input vector and test input vector	τ^*	Nearest training input (knot) to test point $\mathbf{x}^*$
$\mathcal{L}_f, \mathcal{L}_{\text{sp}}$	Lipschitz constant of true function f and spline block	$\mathcal{L}_{\text{MLP}}, \mathcal{L}_{\text{MLP}_i}$	Lipschitz constants of MLP block and the i -th SNR-MLP
$u_{\text{MLP}}, u_{\text{sp}}, u_f$	Error bounds on MLP block, spline block, and target function f	$\mathcal{T}, K$	Input knot matrix ($\mathcal{T} \in \mathbb{R}^{m_k \times d}$) and feature-space knot matrix ($K \in \mathbb{R}^{m_k \times q}$).
ξ, ξ_i	Aggregated latent feature vector before spline mapping and its i -th component	m_k	Number of knots per spline

TABLE 11. Results of test errors and percentage of violations on real datasets *over the first two input features*. **Bold and gray bold italic** values indicate the best and second-best models, respectively. The proposed method achieves **the fewest violations across datasets; RMSE is competitive with, but not uniformly lower than, the best baseline for each dataset.**

Model	RMSE						Violations (%)					
	Concrete	Airfoil Self-Noise	Power Plant	Red Wine	White Wine	Real Estate	Concrete	Airfoil Self-Noise	Power Plant	Red Wine	White Wine	Real Estate
MLP	0.79 ± 0.05	0.79 ± 0.03	0.27 ± 0.01	0.90 ± 0.02	0.97 ± 0.04	0.95 ± 0.12	-	-	-	-	-	-
KAN	0.84 ± 0.05	0.85 ± 0.07	0.30 ± 0.04	0.93 ± 0.04	0.98 ± 0.05	1.04 ± 0.14	-	-	-	-	-	-
GP	0.79 ± 0.04	0.78 ± 0.06	0.27 ± 0.01	0.88 ± 0.03	0.94 ± 0.05	0.90 ± 0.12	0.05 ± 0.15	0.53 ± 0.73	0.87 ± 0.26	0.69 ± 0.27	0.60 ± 0.39	0.84 ± 0.94
DUE	0.98 ± 0.06	0.98 ± 0.05	0.85 ± 0.29	0.96 ± 0.03	0.97 ± 0.05	1.00 ± 0.13	33.16 ± 2.84	32.36 ± 3.28	37.33 ± 4.54	15.69 ± 1.14	27.10 ± 2.57	32.41 ± 5.74
SNGP	0.99 ± 0.07	1.00 ± 0.04	1.00 ± 0.03	0.96 ± 0.03	0.98 ± 0.05	1.00 ± 0.13	76.89 ± 5.28	79.77 ± 6.01	91.78 ± 1.83	94.50 ± 10.82	63.82 ± 15.19	62.77 ± 14.30
DK1	0.80 ± 0.06	0.85 ± 0.03	0.27 ± 0.01	0.89 ± 0.03	1.03 ± 0.19	0.90 ± 0.11	0.00 ± 0.00	1.03 ± 0.69	0.00 ± 0.00	0.38 ± 0.27	2.12 ± 0.99	0.24 ± 0.48
DK2	0.85 ± 0.11	0.86 ± 0.34	0.30 ± 0.05	0.96 ± 0.04	1.00 ± 0.06	1.16 ± 0.11	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
KDK2	0.78 ± 0.05	0.82 ± 0.03	0.27 ± 0.01	0.89 ± 0.02	0.97 ± 0.04	0.98 ± 0.12	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
KDK3	0.90 ± 0.18	0.84 ± 0.04	0.47 ± 0.04	0.89 ± 0.03	0.95 ± 0.05	0.94 ± 0.12	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
ENS2	0.79 ± 0.04	0.77 ± 0.04	0.29 ± 0.02	0.89 ± 0.02	0.97 ± 0.07	0.97 ± 0.11	39.85 ± 10.02	30.33 ± 7.04	30.87 ± 11.83	42.78 ± 7.58	44.90 ± 7.40	28.43 ± 8.07

TABLE 12. Results for high-dimensional experiments across two feature extraction methods, PCA and ResNet-18.

	RMSE ↓							IoU ↑							SDA ↑						
Model	2	5	10	20	50	100	200	2	5	10	20	50	100	200	2	5	10	20	50	100	200
(a) PCA components from images																					
DK1	0.136	0.133	0.133	0.133	0.126	0.129	0.149	0.451	0.456	0.454	0.452	0.469	0.466	0.410	0.676	0.856	0.947	0.967	0.974	0.952	0.969
DK2	0.137	0.139	0.134	0.190	0.123	0.184	0.135	0.448	0.443	0.454	0.375	0.486	0.339	0.451	0.677	0.857	0.947	0.972	0.974	0.950	0.971
KDK2	0.137	0.133	0.130	0.130	0.134	0.135	0.136	0.439	0.450	0.456	0.464	0.444	0.446	0.440	0.726	0.843	0.918	0.929	0.971	0.945	0.978
KDK3	0.137	0.131	0.130	0.128	0.130	0.132	0.133	0.447	0.459	0.464	0.470	0.462	0.456	0.454	0.692	0.741	0.862	0.919	0.917	0.911	0.932
ENS1	0.136	0.133	0.133	0.133	0.126	0.130	0.151	0.450	0.457	0.455	0.451	0.467	0.464	0.407	0.571	0.554	0.568	0.563	0.501	0.431	0.378
ENS2	0.135	0.131	0.131	0.128	0.183	0.265	0.129	0.449	0.456	0.446	0.469	0.311	0.145	0.431	0.527	0.542	0.545	0.530	0.588	0.564	0.525
GP	0.138	0.149	0.137	0.161	0.138	0.137	0.143	0.435	0.400	0.450	0.359	0.437	0.448	0.420	1.000	1.000	1.000	1.000	1.000	1.000	1.000
APXGP	0.138	0.137	0.137	0.137	0.137	0.137	0.137	0.427	0.447	0.447	0.447	0.447	0.447	0.447	0.977	1.000	1.000	1.000	1.000	1.000	1.000
DUE	0.135	0.130	0.130	0.119	0.103	0.104	0.112	0.450	0.470	0.480	0.523	0.554	0.556	0.534	0.682	0.930	0.993	1.000	1.000	1.000	1.000
SNGP	0.138	0.144	0.164	0.155	0.148	0.198	0.207	0.445	0.443	0.421	0.424	0.427	0.244	0.246	0.628	0.840	0.892	0.902	0.867	0.553	0.593
(b) PCA components from ResNet-18 features																					
DK1	0.100	0.097	0.090	0.088	0.079	0.074	0.079	0.535	0.538	0.562	0.557	0.586	0.606	0.589	0.704	0.686	0.875	0.892	0.788	0.837	0.935
DK2	0.100	0.100	0.093	0.103	0.083	0.073	0.072	0.533	0.535	0.553	0.470	0.573	0.608	0.626	0.691	0.707	0.899	0.918	0.858	0.882	0.902
KDK2	0.100	0.095	0.086	0.086	0.077	0.089	0.071	0.536	0.548	0.575	0.575	0.602	0.493	0.640	0.700	0.734	0.819	0.869	0.849	0.860	0.904
KDK3	0.101	0.095	0.086	0.087	0.084	0.090	0.130	0.531	0.546	0.574	0.567	0.553	0.518	0.381	0.690	0.724	0.791	0.830	0.849	0.897	0.908
ENS1	0.100	0.097	0.089	0.089	0.080	0.075	0.081	0.535	0.540	0.565	0.557	0.585	0.603	0.583	0.578	0.551	0.566	0.562	0.364	0.222	0.408
ENS2	0.100	0.094	0.087	0.084	0.174	0.069	0.065	0.530	0.551	0.569	0.575	0.161	0.628	0.646	0.515	0.533	0.573	0.562	0.461	0.550	0.542
GP	0.139	0.140	0.139	0.140	0.140	0.137	0.138	0.431	0.447	0.434	0.427	0.426	0.447	0.444	0.542	1.000	1.000	1.000	1.000	1.000	1.000
APXGP	0.133	0.137	0.137	0.137	0.137	0.137	0.137	0.445	0.447	0.447	0.447	0.447	0.447	0.447	0.729	1.000	1.000	1.000	1.000	1.000	1.000
DUE	0.099	0.095	0.087	0.085	0.076	0.072	0.068	0.537	0.548	0.581	0.575	0.608	0.637	0.650	0.722	0.937	0.990	0.997	1.000	1.000	1.000
SNGP	0.102	0.095	0.107	0.166	0.176	0.204	0.245	0.528	0.547	0.525	0.341	0.305	0.229	0.138	0.591	0.869	0.832	0.544	0.560	0.497	0.512